A COMPARATIVE ANALYSIS OF ROBUST REGRESSION MODELS FOR CARDIOVASCULAR DISEASE MORTALITY

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NOVEMBER 3, 2026
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STAT 311

Table of Contents

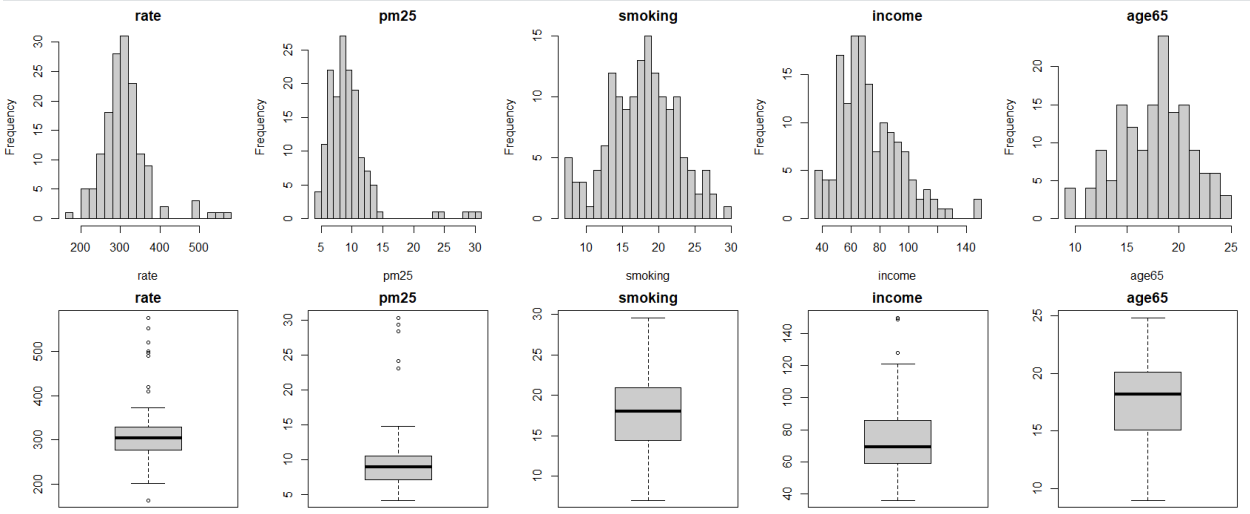
Introduction 2
Exploratory Data Analysis 2
Spread of each variable: SD vs MAD 3
Correlation 3
CVD mortality rate against each predictor 4
OLS Model 4
OLS Diagnostics 5
Model Comparison 6
Model Selection 6
Huber Robust Regression 6
Advantages and Disadvantages Table 7
Conclusion 7
Key Findings 8
Recommendations 8
Appendix 9
Summary statistics 9
Outliers 9
Pearson correlation 9
Spearman correlation 9
Huber Model Output 9
Influential Communities Plot 10
OLS Model Output 10
OLS Confidence Interval 10
Quantile Robust Regression Output 11
Thiel-Sen Regression Output 11
Bi- Square Regression Output 11
R Code 12

Introduction

Cardiovascular disease (CVD) is a leading cause of death, and fine particulate air pollution (PM2.5) is a suspected risk factor. This report uses data from 150 communities to ask how strongly long-term PM2.5 exposure is associated with age-standardized CVD mortality, after adjusting for smoking prevalence, income and the share of residents aged 65 or older. Exploratory analysis shows that a few extreme communities, mainly five with very high PM2.5, can distort ordinary least squares fit. We therefore compare OLS with robust methods (Huber, bi-square, median regression and Theil-Sen with bootstrap) to see whether the PM2.5 estimate holds up.

Exploratory Data Analysis

The data cover 150 communities with no missing values. Distributions are summarized below, and then we'll compare SD against MAD to show where extreme values inflate the spread.



The histograms and boxplots show that the covariates are in good shape, but the response and the main predictor are not.

- CVD mortality rate: roughly bell-shaped around 300 per 100,000, with a long right tail. The boxplot flags nine communities, eight of them high (490 to 577) and one very low (163).
- PM2.5: strongly right-skewed. Most communities fall between 5 and 15 µg/m³, then there is a gap, and five communities sit at 23 to 30.
- Smoking: roughly symmetric, with no outliers.
- Income: mildly right-skewed. The boxplot flags three high values (128, 149 and 150), but they are only a mild tail.
- Age65: roughly symmetric, with no outliers.

  Rate and PM2.5 are the two variables with problem observations, so they are the ones that need robust methods. Smoking and age65 are well behaved.

Spread of each variable: SD vs MAD

Variable	Median	SD	MAD	SD/MAD(times)	Skew
rate	304.1	60.5	38.9	1.55	1.60
pm25	8.9	3.98	2.52	1.58	3.03
smoking	18.0	4.74	4.74	1.00	−0.13
income	68.9	21.1	19.1	1.10	0.94
age65	18.2	3.45	3.48	0.99	−0.25

The SD is about 1.5 times the MAD for rate and pm25, so extreme values are inflating their spread. For smoking, income and age65 the two measures agree, so those variables are clean.

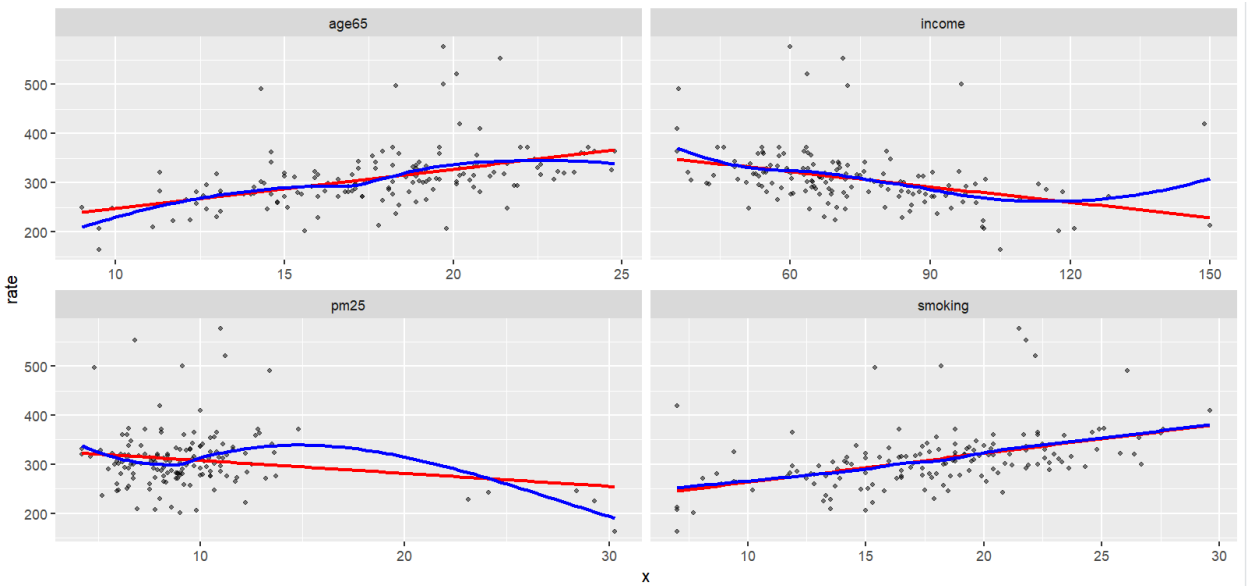
Correlation

	Pearson with rate	Spearman with rate
pm25	−0.17	0.00
smoking	0.46	0.54
income	−0.36	−0.50
insage65	0.46	0.57

	rate	pm25	smoking	income	age65
rate	1.00	-0.17	0.46	-0.36	0.46
pm25	-0.17	1.00	-0.13	0.10	-0.31
smoking	0.46	-0.13	1.00	-0.76	0.00
income	-0.36	0.10	-0.76	1.00	-0.02
age65	0.46	-0.31	0.00	-0.02	1.00

- PM2.5: the Pearson correlation is weakly negative (−0.17), but the rank-based Spearman correlation is zero. The apparent link comes from a few extreme communities, not from the bulk of the data.
- Other covariates: Spearman is stronger than Pearson for smoking, income and age65, so their relationships hold up, and the high-mortality outliers only weaken the linear measure.
- Collinearity: smoking and income are strongly correlated with each other (−0.76), but the VIFs are about 2.4, so collinearity is not a concern.

CVD mortality rate against each predictor



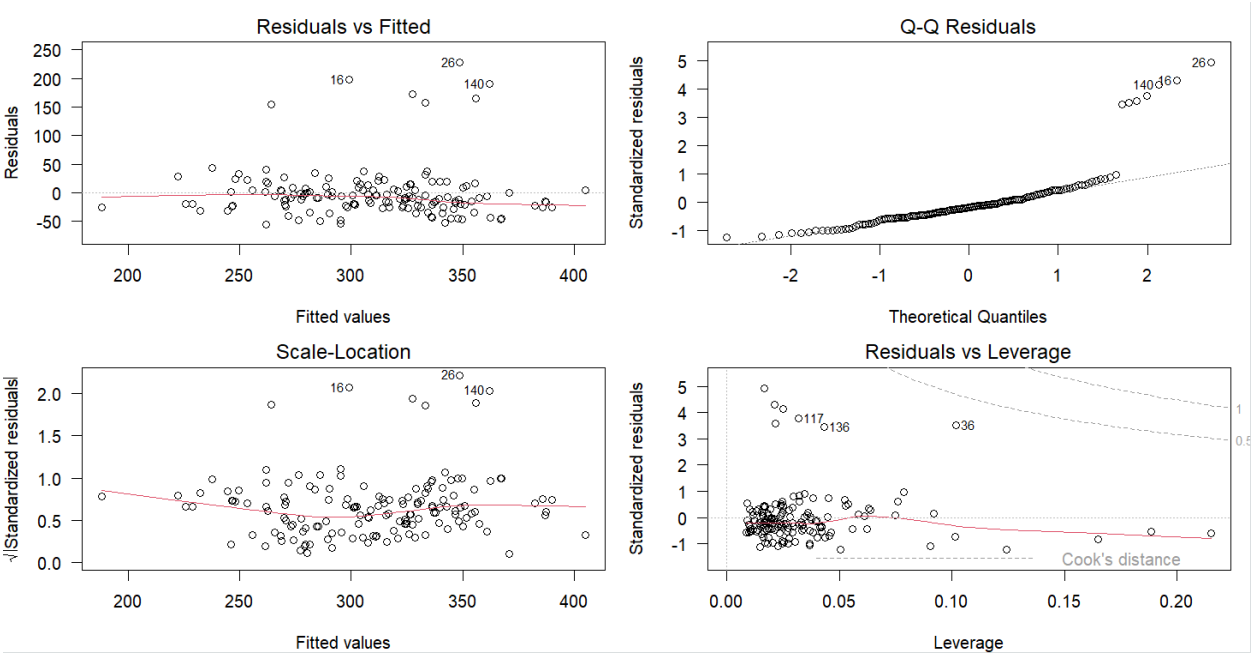
Smoking and age65 rise with mortality, income falls, and in each case the two curves track each other, so those relationships are linear and consistent. The exception is PM2.5, where the OLS line slopes down while the loess curve is flat until PM2.5 reaches about 15, confirming that the downward trend comes from the five high-PM2.5 communities. Income bends upward only above 145, where just two communities sit, so that bend isn't a real pattern. Because rate and pm25 are strongly right-skewed, we also inspected them on the log scale. Both become close to symmetric, which suggests the skew is driven by a short tail of extreme communities rather than by the underlying shape of the variables.

OLS Model

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	61.91332	48.04215	1.289	0.200	
pm25	0.42327	1.01601	0.417	0.678	
smoking	5.78092	1.24219	4.654	7.29e-06	***
income	-0.04175	0.27746	-0.150	0.881	
age65	8.14191	1.16248	7.004	8.62e-11	***

The unadjusted PM2.5 slope is −2.63, suggesting a strong protective effect of air pollution. After adjusting for smoking, income and age65, the slope reverses to 0.42 with (95% CI −1.58 to 2.43). The raw association was therefore driven by confounding, the most polluted communities are also young and have low smoking prevalence. Even after adjustment, however, the PM2.5 estimate is small and not statistically significant, and its wide interval may reflect distortion from the extreme communities identified in the EDA. The model explains 42% of the variation in mortality as seen here, with smoking and age65 highly significant.

OLS Diagnostics



Three OLS assumptions fail. The residuals-versus-fitted and scale-location plots both show variance increasing with the fitted value, so the standard errors from the model are unreliable and a sandwich estimator is needed. The Q-Q plot shows a heavy upper tail, driven by a handful of high-mortality communities, so normal-theory p-values are optimistic. The residuals-versus-leverage plot locates the influential observations: ids 36, 117, 136 and 26 combine large studentized residuals with enough leverage to matter. Only id 36 crosses the 0.5 Cook's D contour, but the remaining points still exceed the sample cook's D cutoff of 0.0267. Together the three findings point in one direction: a few communities are driving the fit, which justifies the robust methods that follow.

Model Comparison

Predictor	OLS	Huber	Quantile	Bi square	Thiel-Sen
Intercept	61.91	131.53	123.75	146.67	299.56
PM2.5	0.42	0.25	1.06	0.23	-0.61
Smoking	5.78	4.08	4.06	3.71	NA
Income	-0.04	-0.4	-0.39	-0.48	NA
Age65	8.14	7.06	7.04	6.85	NA

Several robust regression methods were fitted to evaluate the sensitivity of the results to influential observations and departures from normality. The Huber regression produced a PM2.5 coefficient of 0.25, the quantile regression produced an estimate of 1.06, and the Theil-Sen estimator produced a slope of -0.61 and Bi-square regression produced a PM2.5 coefficient of 0.23. Although the estimated effects varied somewhat across methods, all approaches led to the same substantive conclusion: there was insufficient evidence of an association between PM2.5 exposure and cardiovascular disease mortality after adjustment for smoking prevalence, income, and age structure.

In contrast, smoking prevalence and the proportion of residents aged 65 years or older consistently demonstrated strong positive associations with cardiovascular disease mortality, while income was consistently associated with lower mortality rates. The agreement among the robust and classical methods suggests that the findings are stable and not primarily driven by outliers or influential communities.

Model Selection

Huber Robust Regression

$$\text{Hat}\{\text{CVD Rate}\} = 131.53 + 4.08\text{Smoking} - 0.40\text{Income} + 7.06\text{Age65}$$

The Huber robust regression model was selected as the preferred model because it provides a balance between efficiency and robustness. Unlike OLS, which can be heavily influenced by extreme observations, Huber regression reduces the impact of outliers while still using information from all observations. Compared with quantile regression, which focuses on the conditional median, Huber regression models the conditional mean and therefore remains directly comparable to OLS results. The Theil-Sen estimator is highly robust but can be less efficient when multiple predictors are included. Since the dataset exhibited potential outliers but no evidence that a median-based interpretation was specifically required, Huber regression provided the most appropriate compromise between robustness and interpretability.

Advantages and Disadvantages Table

Method	Advantages	Disadvantages
OLS	Easy to interpret; most efficient when assumptions hold	Sensitive to outliers and influential observations
Huber Robust Regression	Robust to outliers while retaining efficiency	Requires choice of tuning parameter
Quantile Regression	Robust to outliers; models median rather than mean	Interpretation differs from OLS; less efficient when assumptions hold
Theil-Sen Regression	Very robust to extreme values	Computationally intensive and less flexible with multiple predictors
Bi-square Robust Regression	More aggressively downweights extreme outliers; highly robust to influential observations	Interpretation differs from OLS and mean-based models

Conclusion

This analysis examined the relationship between PM2.5 exposure and cardiovascular disease mortality across communities. Exploratory analyses identified several influential observations and evidence of non-normal residuals. However, robust regression approaches produced conclusions similar to those obtained from ordinary least squares regression.

After adjustment for smoking prevalence, income, and the proportion of older adults, PM2.5 was not found to be a statistically significant predictor of cardiovascular disease mortality. Smoking prevalence and population age structure were the strongest predictors of mortality, while higher income was associated with lower mortality rates.

The main trade-off was between efficiency and robustness. OLS is most efficient when assumptions are met but is sensitive to outliers. Huber and Bi-square improve robustness by reducing the influence of extreme observations, with Bi-square providing stronger protection. Quantile regression is robust but changes the focus from the mean to the median. Theil-Sen offers high robustness but may sacrifice efficiency and flexibility. Therefore, the preferred model was selected based on the best balance between robustness, efficiency, and interpretability.

Overall, the consistency of results across OLS, Huber regression, quantile regression, and Theil-Sen estimation and Bi-square regression suggests that the main findings are robust to influential observations and violations of classical regression assumptions.

Key Findings

- **PM2.5 was not a statistically significant predictor** of cardiovascular disease rates across the OLS, Huber robust, Quantile, Theil-Sen, and Bi-square models.
- **Smoking prevalence showed a positive association** with cardiovascular disease rates, indicating that regions with higher smoking rates tended to have higher disease rates.
- **Income was negatively associated** with cardiovascular disease rates, suggesting that higher-income regions generally experienced lower disease burdens.
- **The proportion of the population aged 65 years and older was the strongest positive predictor** of cardiovascular disease rates across all models.
- OLS diagnostic plots revealed heteroscedasticity, non-normal residuals, and influential observations, indicating that classical regression assumptions were violated.
- Across OLS, Huber, Bi-square, Quantile, and Theil-Sen regression models, PM2.5 was not a statistically significant predictor of CVD mortality after adjustment for smoking, income, and age structure.
- **The conclusions were consistent across modeling approaches**, demonstrating that the findings were robust to outliers and different estimation methods.
- **Huber robust regression was selected as the final model** because it provided a practical balance between robustness to outliers and statistical efficiency while maintaining straightforward interpretation.

Recommendations

- **Focus public health interventions on smoking reduction**, as smoking prevalence was consistently associated with higher cardiovascular disease mortality across all models.
- **Prioritize cardiovascular health programs for older populations**, since the proportion of residents aged 65 years and older was the strongest predictor of mortality.
- **Consider socioeconomic factors when designing health policies.** The negative relationship between income and cardiovascular disease mortality suggests that addressing income-related health inequalities may help reduce disease burden
- **Collect additional environmental and health risk factors in future studies.** Variables such as physical activity, obesity, healthcare access, and education level may help explain variation in cardiovascular disease rates that was not captured by the current model.
- **Use robust regression methods when influential observations are present.** The exploratory analysis and diagnostic plots identified outliers and violations of OLS assumptions, demonstrating the value of robust approaches such as Huber and Bi-square regression.

Appendix

Summary statistics

```
> summary(cvd)
      rate            pm25           smoking          income           age65             id
 Min.   :163.1   Min.   : 4.200   Min.   : 7.00   Min.   : 35.80   Min.   : 9.00   Min.   :  1.00
 1st Qu.:277.2   1st Qu.: 7.100   1st Qu.:14.45   1st Qu.: 58.85   1st Qu.:15.15   1st Qu.: 38.25
 Median :304.1   Median : 8.900   Median :18.00   Median : 68.90   Median :18.15   Median : 75.50
 Mean   :309.4   Mean   : 9.407   Mean   :17.75   Mean   : 73.03   Mean   :17.68   Mean   : 75.50
 3rd Qu.:329.5   3rd Qu.:10.500   3rd Qu.:20.88   3rd Qu.: 85.62   3rd Qu.:20.10   3rd Qu.:112.75
 Max.   :576.6   Max.   :30.300   Max.   :29.60   Max.   :150.00   Max.   :24.80   Max.   :150.00
```

Outliers

```
+ }
rate :  9 boxplot outliers -> 163.1 409.9 419.2 490.6 497.1 499.9 520.3 552.9 576.6
pm25 :  5 boxplot outliers -> 23.1 24.1 28.4 29.3 30.3
smoking :  0 boxplot outliers ->
income :  3 boxplot outliers -> 128.2 148.8 150
age65 :  0 boxplot outliers ->
```

Pearson correlation

	rate	pm25	smoking	income	age65
rate	1.00	0.00	0.54	-0.50	0.57
pm25	0.00	1.00	-0.06	0.10	-0.22
smoking	0.54	-0.06	1.00	-0.71	0.00
income	-0.50	0.10	-0.71	1.00	-0.03
age65	0.57	-0.22	0.00	-0.03	1.00

Spearman correlation

	rate	pm25	smoking	income	age65
rate	1.00	0.00	0.54	-0.50	0.57
pm25	0.00	1.00	-0.06	0.10	-0.22
smoking	0.54	-0.06	1.00	-0.71	0.00
income	-0.50	0.10	-0.71	1.00	-0.03
age65	0.57	-0.22	0.00	-0.03	1.00

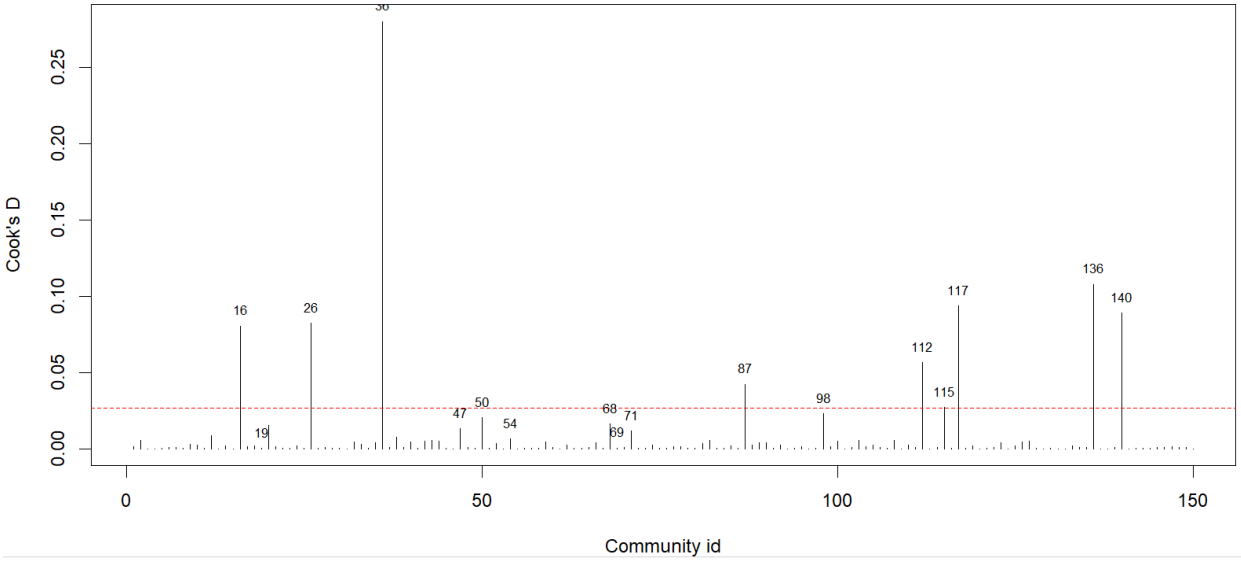
Huber Model Output

```
Call: rlm(formula = rate ~ pm25 + smoking + income + age65, data = cvd,
    psi = psi.huber)
Residuals:
    Min      1Q  Median      3Q     Max
-46.426 -15.286  -0.516  15.067 239.418

Coefficients:
            Value    Std. Error t value
(Intercept) 131.5281  25.4903     5.1599
pm25          0.2541   0.5391     0.4713
smoking       4.0810   0.6591     6.1919
income       -0.4002   0.1472    -2.7188
age65         7.0626   0.6168    11.4505

Residual standard error: 22.46 on 145 degrees of freedom
```

Influential Communities Plot



OLS Model Output

```
Call:
lm(formula = f, data = cvd)

Residuals:
    Min      1Q  Median      3Q     Max
-54.743 -22.442  -8.513   9.007 227.850

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 61.91332   48.04215   1.289    0.200
pm25         0.42327    1.01601   0.417    0.678
smoking      5.78092    1.24219   4.654 7.29e-06 ***
income      -0.04175    0.27746  -0.150    0.881
age65        8.14191    1.16248   7.004 8.62e-11 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 46.6 on 145 degrees of freedom
Multiple R-squared:  0.4222,    Adjusted R-squared:  0.4063
F-statistic: 26.49 on 4 and 145 DF,  p-value: < 2.2e-16
```

OLS Confidence Interval

```
> confint(ols)
                  2.5 %      97.5 %
(Intercept) -33.0400530 156.8666932
pm25         -1.5848275   2.4313584
smoking       3.3257704   8.2360646
income       -0.5901322   0.5066297
age65         5.8443215  10.4395065
>
```

Quantile Robust Regression Output

```
Call: rq(formula = rate ~ pm25 + smoking + income + age65, tau = 0.5,
    data = cvd)

tau: [1] 0.5

Coefficients:
            coefficients lower bd  upper bd
(Intercept) 123.75326     72.22403 177.21875
pm25          1.05710     -1.06471   4.87490
smoking       4.06446      3.00259   5.28325
income       -0.39398     -0.65199  -0.06208
age65         7.03565      6.13043   8.21675
> |
```

Thiel-Sen Regression Output

```
Call:
mblm(formula = rate ~ pm25, dataframe = cvd)

Residuals:
     Min       1Q   Median       3Q      Max
-117.888  -15.842    9.324   35.317  283.779

Coefficients:
            Estimate      MAD V value Pr(>|V|)
(Intercept) 299.5646  61.4622   11212   <2e-16 ***
pm25         -0.6131   6.7545    6125    0.386
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 62.32 on 148 degrees of freedom
```

Bi- Square Regression Output

```
Coefficients:
            Value    Std. Error t value
(Intercept) 146.6752  23.7273     6.1817
pm25          0.2323   0.5018     0.4629
smoking       3.7173   0.6135     6.0592
income       -0.4891   0.1370    -3.5691
age65         6.8487   0.5741    11.9287

Residual standard error: 22.34 on 145 degrees of freedom
> |
```

R Code

```
getwd()

setwd("C:/Users/grace/OneDrive/Documents/School Stuff/Semester 4/Advanced Regression/Lab 4 assignment")

cvd<- read.csv("cvd_mortality.csv")

dim(cvd)

# ---- Setup ----

library(ggplot2)

library(GGally)

library(car)


cvd$community <- NULL

cvd$id <- seq_len(nrow(cvd))


str(cvd)

summary(cvd)

colSums(is.na(cvd))


# ---- 1. Summary table (mean, sd, median, MAD, IQR) ----

vars <- c("rate", "pm25", "smoking", "income", "age65")

eda_tab <- t(sapply(cvd[vars], function(x)

  c(n = length(x), mean = mean(x), sd = sd(x), median = median(x),

    MAD = mad(x), IQR = IQR(x), min = min(x), max = max(x),

    skew = mean((x - mean(x))^3) / sd(x)^3)))

round(eda_tab, 2)


# ---- 2. Distributions: histogram + boxplot for each variable ----

op <- par(mfrow = c(2, 5), mar = c(4, 4, 2, 1))

for (v in vars) hist(cvd[[v]], main = v, xlab = v, col = "grey80", breaks = 20)
```

```
for (v in vars) boxplot(cvd[[v]], main = v, col = "grey80", horizontal = FALSE)
par(op)

# Which values are flagged by the boxplot rule (1.5 * IQR)?
for (v in vars) {
  out <- boxplot.stats(cvd[[v]])$out
  cat(v, ": ", length(out), "boxplot outliers ->", sort(out), "\n")
}

# Is income top-coded?
sum(cvd$income == max(cvd$income))
sort(cvd$income, decreasing = TRUE)[1:10]

# ---- 3. Robust z-scores: rows that are extreme on any variable ----
rz <- sapply(cvd[vars], function(x) (x - median(x)) / mad(x))
flag <- which(apply(abs(rz), 1, max) > 3)
cbind(id = flag, round(cvd[flag, vars], 1), round(rz[flag, ], 1))

# ---- 4. Relationships: scatterplot matrix with correlations ----
ggpairs(cvd[vars],
        lower = list(continuous = wrap("smooth", alpha = 0.4, size = 1)))
round(cor(cvd[vars]), 2)
round(cor(cvd[vars], method = "spearman"), 2)   # compare: big gaps hint at outlier influence

# ---- 5. Main relationship: rate vs pm25 (OLS line vs lowess vs robust-ish) ----
ggplot(cvd, aes(pm25, rate)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, colour = "red") +
  geom_smooth(method = "loess", se = FALSE, colour = "blue") +
```

```
  labs(title = "CVD mortality vs PM2.5",
       subtitle = "red = OLS, blue = loess",
       x = "PM2.5 (µg/m³)", y = "CVD deaths per 100,000")

# Same plot for each covariate in one panel
long <- reshape(cvd[c("id", "rate", "pm25", "smoking", "income", "age65")],
                varying = c("pm25", "smoking", "income", "age65"),
                v.names = "x", timevar = "var",
                times = c("pm25", "smoking", "income", "age65"),
                direction = "long")
ggplot(long, aes(x, rate)) +
  geom_point(alpha = 0.5, size = 1) +
  geom_smooth(method = "lm", se = FALSE, colour = "red") +
  geom_smooth(method = "loess", se = FALSE, colour = "blue") +
  facet_wrap(~ var, scales = "free_x")

# ---- 6. Skew check: does a log transform help? (pm25, income, rate) ----
op <- par(mfrow = c(2, 3))
for (v in c("rate", "pm25", "income")) {
  hist(cvd[[v]], main = v, xlab = v, col = "grey80")
  hist(log(cvd[[v]]), main = paste("log", v), xlab = paste("log", v), col = "grey80")
}
par(op)

# ---- 7. Multicollinearity between the predictors ----
vif(lm(rate ~ pm25 + smoking + income + age65, data = cvd))

# ---- 8. Quick look at the suspicious pm25 points ----
cvd[order(-cvd$pm25)[1:8], ]
```

```
cvd[order(-cvd$rate)[1:8], ]


library(MASS)
library(car)
library(lmtest)
library(sandwich)
library(ggplot2)

cvd <- read.csv("cvd_mortality.csv")
cvd$id <- seq_len(nrow(cvd))
f <- rate ~ pm25 + smoking + income + age65

# ---- 1. Fit OLS ----
ols <- lm(f, data = cvd)
summary(ols)
confint(ols)

# Unadjusted vs adjusted PM2.5 slope (how much confounding there is)
cat("unadjusted pm25 slope:", coef(lm(rate ~ pm25, data = cvd))["pm25"], "\n")
cat("adjusted  pm25 slope:", coef(ols)["pm25"], "\n")

# ---- 2. Standard four-panel diagnostics ----
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1)); plot(ols, las = 1); par(mfrow = c(1, 1))

# ---- 3. Influence measures in one table ----
library(MASS)
n <- nrow(cvd); p <- length(coef(ols))
diag_tab <- data.frame(
  id      = cvd$id,
```

```
  rate     = cvd$rate, pm25 = cvd$pm25, smoking = cvd$smoking,
  income   = cvd$income, age65 = cvd$age65,
  leverage = hatvalues(ols),
  stud_res = studres(ols),
  cooks    = cooks.distance(ols),
  dffits   = dffits(ols)
)

lev_cut <- 2 * p / n         # leverage cutoff
cd_cut  <- 4 / n             # Cook's D cutoff
cat("Leverage cutoff:", round(lev_cut, 3),
    " Cook's D cutoff:", round(cd_cut, 4), "\n")

flag <- subset(diag_tab, leverage > lev_cut | abs(stud_res) > 3 | cooks > cd_cut)
flag[order(-flag$cooks), ]

# ---- 4. Residuals vs leverage, labelled + Cook's D index plot ----
ggplot(diag_tab, aes(leverage, stud_res)) +
  geom_point(aes(size = cooks), alpha = 0.5) +
  geom_hline(yintercept = c(-3, 3), linetype = 2) +
  geom_vline(xintercept = lev_cut, linetype = 2) +
  geom_text(data = flag, aes(label = id), vjust = -1, size = 3) +
  labs(x = "Leverage", y = "Studentized residual", size = "Cook's D")

plot(diag_tab$cooks, type = "h", ylab = "Cook's D", xlab = "Community id")
abline(h = cd_cut, col = "red", lty = 2)
text(flag$id, flag$cooks, labels = flag$id, pos = 3, cex = 0.7)
```

```
# ---- 6. Assumption tests ----
bptest(ols)                 # heteroscedasticity
shapiro.test(resid(ols))    # normality of residuals
vif(ols)                    # collinearity
car::ncvTest(ols)




# ---- 8. Sandwich (HC3) standard errors for the OLS fit ----
coeftest(ols, vcov = vcovHC(ols, type = "HC3"))

# ---- 9. Sensitivity: refit without the five high-PM2.5 communities ----
hi_pm <- cvd$pm25 > 20
ols_trim <- lm(f, data = cvd[!hi_pm, ])
summary(ols_trim)

## Huber model
huber_model <- rlm(
  rate ~ pm25 + smoking + income + age65,
  data = cvd,
  psi = psi.huber
)

summary(huber_model)

##Quantile regression
library(quantreg)
```

```
rq_model <- rq(
  rate ~ pm25 + smoking + income + age65,
  tau = 0.5,
  data = cvd
)

summary(rq_model)

## Thiel-Sen Estimation
install.packages("mblm")
library(mblm)

theilsen <- mblm(rate ~ pm25, data = cvd)

summary(theilsen)

bisquare_model <- rlm(
  rate ~ pm25 + smoking + income + age65,
  data = cvd,
  psi = psi.bisquare
)

summary(bisquare_model)
```